RELATIVELY PERFECT COMPLEXES FORM A SATURATED TRIANGULATED SUBCATEGORY FOR FLAT FINITELY PRESENTED MAPS OF SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 12366-12373 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 12375-12381, the complete author proof (5 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. BORDERLINE for Chief: proof is an enumeration of four closure lemmas with the assembly implicit; accepted under the written-reduction rule (each axiom's discharge is named one-for-one), flagged if Chief prefers to reject pure enumeration cards. The lane's definition-verification style treats it like the fpqc-local stubs family (compare L16080).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Definition of Y-perfect (relative pseudo-coherent plus finite tor amplitude, chapter definition section); Derived Categories, definition saturated; the four Cohomology on Sites closure lemmas.

Source reference. Original lemma 05RB: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12381–12381. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08FV: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12381–12381. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08G2: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12381–12381. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08FW: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12381–12381. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 08G3: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12381–12381. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S which is flat and locally of finite presentation. The full subcategory of $D(\mathcal{O}_X)$ consisting of Y-perfect objects is a saturated*[1] *triangulated subcategory.*

Proof. This follows from Cohomology on Sites, Lemmas 08FV, 08FW, 08G2, and 08G3. □

[1] Derived Categories, Definition 05RB.